



**RECEP TAYYİP ERDOĞAN UNIVERSITY
FACULTY OF ENGINEERING AND ARCHITECTURE
COMPUTER ENGINEERING DEPARTMENT**

CEN207 Data Structures

2025-2026 Fall

Homework No: 49

Due Date: 01.01.2026

Team Members:

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2026

PROJECT REPORT: INVENTORY MANAGEMENT APPLICATION

1. PROJECT OVERVIEW

This project implements a comprehensive Inventory Management Application using Java. The system provides functionality for managing materials, projects, and expenses with advanced data structures and algorithms integration.

The application consists of three core modules:

- **Material Management Module:** Complete CRUD operations with undo/redo functionality
- **Project Tracking Module:** Project lifecycle management with dependency tracking and cycle detection
- **Expense Logging Module:** Expense tracking and reporting associated with projects and materials

2. PROJECT GOALS

- Develop a comprehensive system for managing materials, projects, and expenses
- Implement advanced data structures (B+ Trees, Hash Tables) for efficient data storage
- Apply algorithms (Huffman Coding, KMP, Graph algorithms) for optimized operations
- Achieve high test and documentation coverage (95%+)

3. PROBLEMS TO SOLVE

- Need for efficiency and speed in inventory management
- Management of project dependencies and cycle detection
- Optimized search and access for large datasets
- Cost tracking and reporting needs
- Reliable and consistent data storage

4. TECHNICAL IMPLEMENTATION

The system implements 12 different algorithms and data structures:

Data Structures:

- **B+ Trees:** Efficient indexing and range queries for large datasets
- **Hash Tables:** Fast key-value lookups with $O(1)$ average time complexity
- **Double Linked Lists:** Material history tracking

Algorithms:

- **Graph Algorithms (BFS, DFS):** Project dependency traversal
- **Strongly Connected Components:** Cycle detection in project dependencies
- **KMP Algorithm:** Efficient string pattern matching for search operations
- **Huffman Coding:** Data compression for backup operations
- **File Operations:** Atomic file writes with hash-based integrity checks

5. TEAM CONTRIBUTIONS

- **Harun Akçay:** Project Tracking and Expense Logging modules, Huffman Coding algorithm, 50+ test cases
- **Sudenaz Orhan:** Material Inventory module, HashTable, Double Linked List, BFS/DFS/SCC/KMP algorithms
- **Zümre Uykun:** Console menu, File Operations algorithm, Sales Tracker module, documentation improvements
- **Güler Dinç:** Sales Tracker module, code coverage improvements, build scripts, Doxygen fixes

6. RESULTS AND ACHIEVEMENTS

- 100% completion of all three core modules
- 12 algorithms and data structures fully implemented and integrated
- 93.27% test coverage with 716 test cases
- 97.8% documentation coverage
- Comprehensive Maven site documentation with Doxygen API docs, JaCoCo coverage reports, and Coverxygen documentation coverage

7. TECHNOLOGY STACK

- Java 8
- Maven for build management
- JUnit 4 for testing
- Doxygen for API documentation
- JaCoCo for test coverage
- Coverxygen for documentation coverage
- Repository and Service design patterns

8. CONCLUSION

This project successfully demonstrates the integration of advanced computer science concepts into a practical, real-world application. The high test and documentation coverage ensure maintainability and reliability. The modular architecture allows algorithms to be used across different modules, demonstrating the power of well-designed software architecture.